

Continual Test-Time Training on Graphs via Adaptive Prompts Integration

Qianyi Cai , Ziyue Qiao , *Member, IEEE*, Rui Cai , Huijie Liu , Junyi Li , Xiao Luo ,
and Hui Xiong , *Fellow, IEEE*

Abstract—This paper investigates a novel and critical problem of *Graph Continual Test-Time Training*, which aims to enable a frozen pre-trained graph model to adapt continuously to evolving out-of-distribution (OOD) graphs without supervision. Existing test-time training methods primarily focus on one-step adaptation and overlook long-term knowledge retention, while conventional continual learning approaches rely on labeled data and static memory replay. Consequently, they are unable to handle sequential OOD domains effectively, often suffering from severe forgetting and limited efficiency in dynamic graph environments. To address these challenges, we propose DPCGL (Dynamic Prompts-based Continual Graph Learning), a data-centric framework that performs continual test-time training through adaptive prompt optimization. DPCGL freezes the pre-trained backbone and maintains a dynamic prompt pool, where prompts are adaptively selected and updated for each incoming graph domain. This design enables parameter-efficient adaptation and mitigates forgetting by organizing transferable knowledge within prompts rather than model weights. Furthermore, DPCGL jointly optimizes three objectives: similarity alignment for representativeness, KL divergence regularization for knowledge preservation, and diversity constraint for generalization, providing both stability and adaptability during continual adaptation. Extensive experiments on multiple evolving OOD graph benchmarks demonstrate that DPCGL achieves state-of-the-art performance, effectively alleviating catastrophic forgetting and enabling robust continual adaptation across domains.

Index Terms—Continual learning, domain adaptation, graph neural networks.

I. INTRODUCTION

GRAPHS are ubiquitous in real-world systems and have been extensively applied in diverse areas such as data mining, knowledge discovery, and other tasks across different domains. As a powerful representation paradigm, classical Graph Neural Network (GNN) models, such as Graph Convolutional Networks (GCN) [1], GraphSAGE [2], and Graph Attention Networks (GAT) [3], have demonstrated significant success due to their ability to capture complex structural relationships. These models have been effectively applied in fields such as social networks [4], [5], computational biology [6], behavioral analysis [7] and recommendation systems [8].

However, as graph data continuously evolves across diverse domains, Out-of-Distribution (OOD) problems naturally arise, where graph distributions gradually deviate from the initial independent and identically distributed (i.i.d.) settings assumed in the model training process. A model trained on earlier graphs often struggles to generalize effectively when applied to newly observed graphs, leading to noticeable performance degradation. As shown in Fig. 1, a GNN trained on one region's social network may perform poorly in another due to domain discrepancy. Fig. 2 further visualizes the embeddings of class-0 and class-1 nodes across evolving OOD Twitch datasets, clearly illustrating such domain shifts. This raises a key question: *How can a pre-trained GNN model continuously adapt to evolving OOD graphs while avoiding catastrophic forgetting and excessive retraining?*

Test-time training (TTT) methods have been widely studied to improve model robustness against OOD graphs [9], [10], [11]. While effective for one-step adaptation, they are less suited to dynamic settings where graphs arrive sequentially. On one hand, training an independent model for each new graph is inefficient in both parameters and storage, and complicates model selection at inference. On the other hand, continuously fine-tuning a single model over successive graphs tends to overwrite earlier adaptations, causing catastrophic forgetting. These limitations reveal the need for a framework that can adapt to a sequence of OOD graphs in a continual manner while avoiding repeated retraining of model parameters.

A natural direction to mitigate forgetting is continual learning (CL) [12], [13], [14], [15], which has been extensively explored in supervised scenarios. However, applying CL to the test-time setting is far from straightforward. Traditional

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Qianyi Cai and Junyi Li are with the Thrust of Artificial Intelligence, The Hong Kong University of Science and Technology (Guangzhou), Guangzhou 511458, China (e-mail: qcai603@connect.hkust-gz.edu.cn; jli000@connect.hkust-gz.edu.cn).

Ziyue Qiao is with the School of Computing and Information Technology, Great Bay University, Dongguan 523000, China (e-mail: ziyuejoe@gmail.com).

Rui Cai is with the University of California, Davis, Davis, CA 95616 USA (e-mail: ruicai@ucdavis.edu).

Huijie Liu is with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen 518060, China (e-mail: huijieliu@szu.edu.cn).

Xiao Luo is with the Department of Statistics, University of Wisconsin–Madison, Madison, WI 53706 USA (e-mail: xiao.luo@wisc.edu).

Hui Xiong is with the Thrust of Artificial Intelligence, The Hong Kong University of Science and Technology (Guangzhou), Guangzhou 511458, China, and also with the Department of Computer Science and Engineering, The Hong Kong University of Science and Technology, Hong Kong, SAR, China (e-mail: xionghui@ust.hk).

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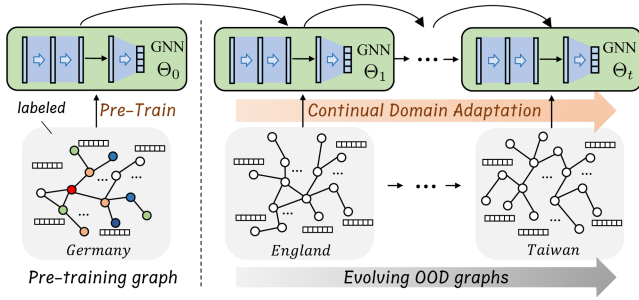


Fig. 1. Problem Illustration: Sequence of continual test-time adaptation on evolving OOD graphs.

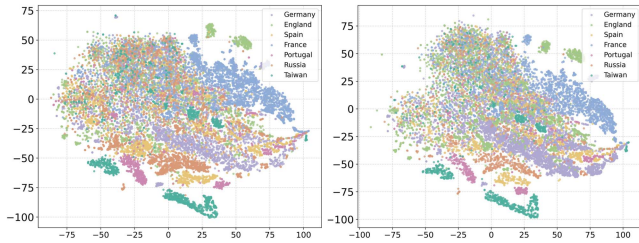


Fig. 2. Embedding visualization of class-0 and class-1 nodes across evolving OOD graphs in Twitch datasets, clearly illustrating domain shifts.

CL methods rely on labeled data, explicit task boundaries, or memory replay buffers, all of which are typically unavailable in unsupervised test-time scenarios. Consequently, directly transferring existing CL techniques to label-free OOD adaptation introduces two fundamental challenges. First, retaining useful knowledge from past domains without supervision or explicit replay is difficult, since determining which representations are task-relevant becomes ambiguous; moreover, replay-based approaches often depend on large, static buffers that are inflexible for dynamically growing graph streams. Second, ensuring stable generalization during continual adaptation is challenging when repeatedly fine-tuning the same parameters, as this easily leads to performance drift and loss of previously acquired generalization ability.

To tackle these challenges, we introduce a data-centric approach that enables continual test-time training via efficient and adaptive prompt optimization. With the development of large language models [16], [17], prompt tuning has become a widely adopted paradigm for parameter-efficient adaptation [18]. Instead of modifying the parameters of the pre-trained GNN, our approach maintains a trainable prompt pool and dynamically selects and refines prompts for each new graph. The motivation is that prompts can serve as efficient, transferable carriers of domain-specific knowledge that can be updated without altering the frozen backbone [19]. This framework enables flexible adaptation to novel domains while maintaining the retention of previously knowledge. In addition, prompt updates provide a compact way to accumulate and organize information across domains, removing the need for large memory buffers and supporting stable, label-free adaptation over time. Compared with replay-based methods, this approach offers a more efficient and scalable pathway for continual graph learning.

In summary, this work introduces the **Dynamic Prompts-based Continual Graph Learning (DPCGL)** framework, which enables label-free, parameter-efficient continual adaptation on evolving OOD graphs. DPCGL incorporates a dynamic prompt pool with adaptive prompt selection, allowing effective test-time training under domain shifts while keeping the backbone frozen. We provide both theoretical justification and empirical evidence demonstrating that DPCGL surpasses current state-of-the-art approaches on multiple graph datasets. The major contributions of this paper are outlined below:

- We propose a novel Dynamic Prompts-based Continual Graph Learning (**DPCGL**) framework for graph continual test-time training, effectively addressing the challenges of continual domain adaptation to evolving OOD graphs.
- We develop a dynamic prompt pool with an adaptive prompt mechanism, which freezes the model while flexibly adjusting prompts to facilitate continual test-time training and improve generalization.
- We conduct extensive theoretical and empirical evaluations on graph datasets, verifying that DPCGL consistently achieves superior and stable improvements over previous state-of-the-art methods across diverse domain conditions.

II. RELATED WORKS

Continual Graph Learning: Existing approaches to continual graph learning (CGL) can generally be grouped into three paradigms: memory replay, parameter regularization and parameter isolation. Memory replay methods mitigate forgetting by preserving a limited memory of representative samples from prior tasks, which are periodically replayed during the learning of new domains [12], [20], [21], [22]. The regularization based methods focuses on stabilizing crucial parameters and important topologies by minimizing changes critical to previous tasks [23], [24], [25], [26], [27]. Parameter isolation approaches allocate new parameters for upcoming tasks while retaining those essential for prior tasks [28]. Our work introduces a novel prompt-based GCL approach, providing a fresh perspective on the field.

Graph Domain Adaptation: Graph domain adaptation has been widely studied. Some methods focus on minimizing domain discrepancy, such as MMD [29] and GRADE [30], while others, like UDAGCN [31], AdaGCN [32] and SGDA [33], are based on adversarial approaches. These are model-centric methods. In contrast, GraphAlign [34] tackles the problem from a data-centric perspective, which aligns with our approach. Beyond source-accessible settings, source-free unsupervised graph adaptation has received increasing attention, where target adaptation proceeds without labeled source graphs; recent findings further highlight *structural/topology mismatch* as a key driver of degradation. Along this line, GTrans [35], GraphCTA [36], GraphATA [37], and Matcha [38] explicitly target graph structural shifts by modeling topology-induced mismatch, which also motivates our protocol-aligned baseline selection for evaluating robustness under structural shift. In parallel, continual test-time adaptation has attracted significant attention in computer vision. CoTTA [39] preserves knowledge from the source model by

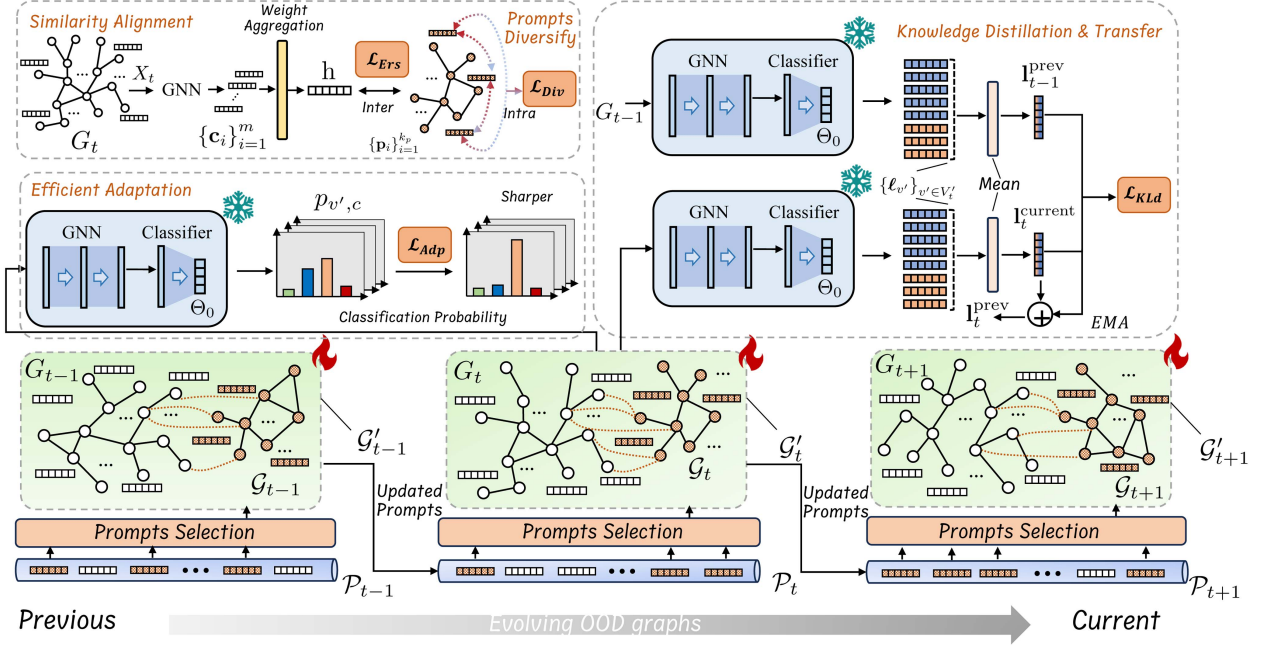


Fig. 3. The DPCGL framework initializes a prompt graph using prompt selection and edge connection mechanisms (Details in III-B) while keeping $f(\Theta_0)$ frozen. The combined graph $\mathcal{G}'_t$ —comprising the input graph G_t and the prompt graph $\mathcal{G}_t$ is used for test-time training. The training employs adaptation loss $\mathcal{L}_{Adp}$ and three losses to mitigate forgetting: representativeness loss $\mathcal{L}_{Ers}$, KL divergence loss $\mathcal{L}_{KLd}$, and diversity loss $\mathcal{L}_{Div}$. Updating the top- k_p prompt parameters $\mathcal{P}_t$ enables continual knowledge accumulation across multiple domains over time.

combining weight-averaged pseudo-labels with averaged predictions. EATA [40] proposes a sample-efficient entropy minimization strategy, optimizing only reliable and non-redundant samples. EcoTTA [41] further improves memory efficiency via self-distilled regularization, while BECoTTA [42] mitigates forgetting via input-dependent online blending of experts under non-stationary test-time shifts. Adaptive representation alignment [43] complements these by studying non-stationary TTA from a representativeness perspective. However, there is limited research on continual domain adaptation on graphs, especially in the unsupervised setting. Our work addresses this gap by exploring novel approaches in this area.

Graph Prompt Learning: Graph prompt learning adapts the concept of prompts from NLP to graphs, offering new insights for graph-related tasks. GPPT [44] introduces graph prompts as additional tokens, incorporating both task and structure information to align node classification and link prediction tasks. GraphPrompt [45] embeds prompts within the hidden layers of graph models, primarily to assist graph pooling operations. All-in-One [46] proposes a general prompt for various downstream tasks, utilizing a learnable subgraph and pretraining via graph-level contrastive learning, while GPF-Plus [47] applies similar data graphs in [46] and extends the concept of prompt tokens to multiple independent basis vectors. PRODIGY [48] employs a hand-crafted prompt graph comprising data graphs and a task graph. TPP [49] introduces prompts to continual graph learning, but in a supervised, class-incremental setting, different from our approach. Guided by a data-centric perspective, our method refines graph prompt mechanisms to align with the Continual Test-Time Training setting, thereby enhancing adaptability across diverse domains.

III. METHODOLOGY

A. Problem Formulation and Overview

Let $f(\Theta_0) : G_s \rightarrow \mathbb{R}^C$ represent a GNN model that has been pre-trained for node classification tasks, where Θ_0 denotes the frozen backbone parameters, G_s refers to the source graph, and C is the number of classes with label set $\mathcal{Y} = \{y_1, y_2, \dots, y_C\}$. We then define a sequence of m out-of-distribution (OOD) target domains $\{G_1, G_2, \dots, G_m\}$, where each graph $G_t = \{V_t, A_t, X_t\}$ corresponds to the t -th domain. Here, V_t denotes the node set, $A_t \in \mathbb{R}^{N_t \times N_t}$ represents the adjacency structure, and $X_t \in \mathbb{R}^{N_t \times d_x}$ is the node attribute matrix with feature dimension d_x . The frozen backbone induces node embeddings in a hidden space of dimension d_h , which we use for prompt retrieval and representativeness alignment. For clarity, we summarize the main notations used throughout the methodology in Table I. Each target graph in the sequence arrives over time and may differ in its data distribution from the others, reflecting the presence of dynamic domain shifts, i.e., $D(G_i) \neq D(G_j)$ for $i \neq j$, where $D(\cdot)$ denotes the underlying distribution. In this framework, the parameters of the pre-trained backbone Θ_0 are kept fixed, while a learnable graph prompt pool $\mathcal{P}$ is introduced to support progressive adaptation to newly encountered domains. At each adaptation step t , we initially select a small subset of prompts $\mathcal{P}_t$ by similarities from $\mathcal{P}$ and integrate it into the target graph G_t to form an augmented graph $\mathcal{G}'_t$. This augmented graph is then used for test-time training. The details of our framework are illustrated in Fig. 3.

B. Dynamic Prompt Graph Design

The number of prompts in the pool remains constant, while their parameters are tunable vertically, allowing training. The

TABLE I
NOTATION SUMMARY IN DPCGL

Symbol	Description
$f(\Theta_0)$	Frozen pre-trained GNN backbone with parameters Θ_0
G_t	Target graph (domain) at step t
V'_t	Node set of the augmented graph $\mathcal{G}'_t$
N'_t	Number of nodes in $\mathcal{G}'_t$ ($N'_t = N_t + k_p$)
$A_t \in \mathbb{R}^{N_t \times N_t}$	Adjacency matrix (graph structure) of G_t
$X_t \in \mathbb{R}^{N_t \times d_x}$	Node feature matrix of G_t (d_x -dim attributes)
$Z_t \in \mathbb{R}^{N_t \times d_h}$	Backbone-induced node embeddings of G_t (used for retrieval and clustering)
$\mathcal{P}$	Global prompt pool
$ \mathcal{P} $	Prompt pool size
k_p	Number of selected prompts at step t
γ	Prompt selection ratio, $k_p = \lceil N_t \gamma \rceil$
$\mathcal{P}_t = \{\mathbf{p}_1, \dots, \mathbf{p}_{k_p}\}$	Selected prompts for step t (trainable in $\mathbb{R}^{d_h}$)
$\mathcal{P}'_t = \phi(\mathcal{P}_t)$	Injected prompt node attributes after projection $\phi: \mathbb{R}^{d_h} \rightarrow \mathbb{R}^{d_x}$
$\mathcal{E}_t$	Edge set of the prompt graph at step t (prompt–prompt edges and prompt–node edges)
$\mathcal{G}_t = (\mathcal{P}'_t, \mathcal{E}_t)$	Prompt graph constructed at step t
$\mathcal{G}'_t$	Augmented graph $G_t \cup \mathcal{G}_t$
$\mathbf{S}$	Similarity matrix for prompts and nodes
τ_s	Similarity threshold for prompt–prompt edges
k_n	Prompt–node top- k_n connections per prompt
$\ell_{v'} \in \mathbb{R}^C$	Output logits of node v'
C	Number of classes
T	Softmax temperature in adaptation/distillation
$\mathcal{L}_{Adp}$	Entropy minimization adaptation loss
$\mathcal{L}_{Ers}$	Representativeness alignment loss
$\mathcal{L}_{Div}$	Diversity regularization loss
$\mathcal{L}_{KLd}$	KL-based knowledge preservation loss (EMA distillation)
$\lambda_1, \lambda_2, \lambda_3$	Loss weights for $\mathcal{L}_{Ers}, \mathcal{L}_{Div}, \mathcal{L}_{KLd}$
α	EMA decay rate used for logit distillation update
m	Number of target domains / adaptation steps in the stream
M	Number of K-means clusters in representativeness alignment

prompt pool length is denoted by $|\mathcal{P}|$, and $\mathcal{P}$ is initialized using the Kaiming uniform distribution. During adaptation, for the t -th target domain, we initially select the top k_p prompts by similarity from the pool, denoted as $\mathcal{P}_t = \{\mathbf{p}_1, \dots, \mathbf{p}_{k_p}\}$. Specifically, we get node embeddings $Z_t \in \mathbb{R}^{N_t \times d_h}$ for G_t through the frozen backbone, and obtain a domain-level reference embedding $\mathbf{z}_t \in \mathbb{R}^{d_h}$ via mean pooling, i.e., $\mathbf{z}_t = \text{MeanPool}(Z_t)$. We then compute the cosine similarity between $\mathbf{z}_t$ and each prompt $\mathbf{p}_j \in \mathbb{R}^{d_h}$ in the global pool $\mathcal{P}$ (initialized by Kaiming uniform) as $s_{t,j} = \cos(\mathbf{z}_t, \mathbf{p}_j)$, rank all prompts by $s_{t,j}$, and retrieve the top- k_p prompts to form $\mathcal{P}_t$ at the beginning of step t , before performing test-time adaptation on G_t . The number of selected prompts is determined as $k_p = \lceil N_t \times \gamma \rceil$, where N_t denotes the total number of nodes in the target domain graph G_t , and γ serves as a scaling factor that controls the prompt selection ratio. In practice, k_p is set to be significantly smaller than N_t to ensure parameter efficiency. When each target domain graph G_t arrives, we construct an associated prompt graph, denoted as $\mathcal{G}_t = (\mathcal{P}'_t, \mathcal{E}_t)$ where $\mathcal{E}_t$ are reassessed based on a cross prompts threshold which is discussed in detail below. As the node-attribute dimension d_x required by the frozen backbone and the prompt embedding dimension d_h are not aligned, we map the selected prompts from the backbone-induced embedding space to the node-attribute space via a differentiable projection

$\phi: \mathbb{R}^{d_h} \rightarrow \mathbb{R}^{d_x}$, i.e., $\mathcal{P}'_t = \phi(\mathcal{P}_t)$, so that each prompt can be injected as an additional node with valid input attributes. Then with the node-prompt threshold, the augmented graph $\mathcal{G}'_t = \{V'_t, A'_t, X'_t\}$ is constructed with the combination of G_t and $\mathcal{G}_t$. Here, V'_t contains the original nodes and the corresponding k_p prompts, $X'_t \in \mathbb{R}^{N'_t \times d_x}$ represents the attribute matrix, where $N'_t = N_t + k_p$ is the number of augmented graph nodes.

1) *Edge Connection*: We construct prompt–prompt and prompt–node edges in the backbone-induced embedding space $\mathbb{R}^{d_h}$, so that prompt selection and connectivity are defined in a consistent similarity space. Specifically, the connections between the prompts $\mathcal{E}_t$ are determined based on the similarity between prompts exceeding a predefined threshold. We first normalize the prompt representations and compute the similarity matrix $\mathbf{S}$ for all prompts and nodes:

$$\mathbf{S} = \text{normalize}(\mathcal{P}_t) \left(\text{normalize} \left(\begin{bmatrix} \mathcal{P}_t \\ Z_t \end{bmatrix} \right) \right)^\top. \quad (1)$$

where *normalize* represents L2 normalization applied to each prompt or node vector. The resulting matrix $\mathbf{S} \in \mathbb{R}^{k_p \times (k_p + N_t)}$ uses its first k_p columns for prompt–prompt similarities and its last N_t columns for prompt–node similarities (corresponding to nodes in V_t). Edges between prompts are established if their similarity exceeds a threshold τ_s , and connections between prompts and nodes are made by selecting the top- k_n most similar nodes:

$$\begin{aligned} \mathcal{E}_t = & \left\{ (i, j) \mid S_{ij} > \tau_s, i, j \in \{1, \dots, k_p\} \right\} \\ & \cup \left\{ (i, j) \mid (i, j) \in \{1, \dots, k_p\} \times \{1, \dots, N_t\}, \right. \\ & \left. k_p + j \in \text{top-}k_n(i) \right\}. \end{aligned} \quad (2)$$

Here, S_{ij} denotes the similarity between prompt i and the j -th column item in $[\mathcal{P}_t; Z_t]$ (i.e., a prompt when $j \leq k_p$ and a node when $j > k_p$). Detailed parameter analysis of top- k parameters is in Fig. 10.

2) *Training and Inference Process*: During the test-time training stage, we dynamically adapt the prompt parameters from $\mathcal{P}_t$ to accommodate incoming target domain graphs. If the shared prompt pool option is enabled, the updated prompt set $\mathcal{P}_t$ is synchronized with the global prompt pool $\mathcal{P}$ *once per domain*, i.e., after finishing adaptation on each incoming graph G_t (before proceeding to G_{t+1}), via a write-back update rule that overwrites only the selected prompts: for the selected index set $\mathcal{I}_t$, we update $\mathbf{p}_i \leftarrow \mathbf{p}_i^{(t)}$ for all $i \in \mathcal{I}_t$, while keeping all unselected prompts unchanged. This synchronization mechanism facilitates continual information accumulation, as prompts incrementally learn and integrate transferable knowledge from multiple domains over time. For the current domain, the training-augmented graph $\mathcal{G}'_t$ is then used as input for inference with $f(\Theta_0)$, ensuring consistency between adaptation and prediction. For previously encountered target graphs, inference is performed using the top- k_p prompts selected from the latest updated dynamic prompt pool, thereby enabling adaptive knowledge reuse across domains.

C. Efficient Adaptation Via Dynamic Prompts

At the t -th adaptation stage, our goal is to improve the model's predictive performance on the augmented graph $\mathcal{G}'_t$ by optimizing the prompt parameters $\mathcal{P}_t$ while keeping the pre-trained backbone Θ_0 fixed. Instead of relying on traditional model parameter adjustments, we adopt a data-centric approach by refining prompt inputs to maximize the information extracted from the data [33], [50]. This adaptation process integrates an adaptive prompt graph $\mathcal{G}_t$ into the input graph G_t , thus ultimately refining the final predictions of the frozen model. The adaptation setting is unsupervised, and we employ an entropy minimization strategy to fine-tune the prompt parameters $\mathcal{P}_t$. Specifically, for the augmented graph $\mathcal{G}'_t$, we obtain the output logits $\ell_{v'} \in \mathbb{R}^C$ from the frozen backbone $f(\Theta_0)$ and compute the softmax probabilities for each node $v' \in V'_t$ and class $c = 1, \dots, C$ as $p_{v',c} = \frac{\exp(\ell_{v',c}/T)}{\sum_{c'=1}^C \exp(\ell_{v',c'}/T)}$, where $T > 0$ is a temperature parameter that controls the sharpness of the output distribution. The adaptation loss $\mathcal{L}_{Adp}$ is then defined as the average entropy of the predictions:

$$\mathcal{L}_{Adp} = -\frac{1}{N'_t} \sum_{v' \in V'_t} \sum_{c=1}^C p_{v',c} \log(p_{v',c} + \epsilon), \quad (3)$$

where ϵ is a small constant introduced to prevent numerical instability during the calculation. Minimizing $\mathcal{L}_{Adp}$ encourages the fixed model to produce sharper, more decisive predictions, thereby reducing uncertainty in the probability outputs and stabilizing adaptation on $\mathcal{G}'_t$. Rather than modifying the model parameters, we instead update the prompts in a data-centric manner to capture the new information in G_t . To further illustrate the intuition behind entropy minimization, reducing $\mathcal{L}_{Adp}$ essentially drives the model to assign higher probabilities to specific classes for each node, resulting in sharper and more decisive output distributions. Such an approach proves particularly effective in unsupervised settings where labels are unavailable, as it naturally guides the model toward confident decision-making without explicit supervision and enhances its stability during continual adaptation.

D. Leveraging Strategies for Forgetting Mitigation

To achieve adaptation and mitigate forgetting efficiently, we leverage multiple loss components. While adapting to the new graph G_t , it is essential to ensure that the prompts remain representative of the current data while alleviating the forgetting of previously acquired knowledge. To achieve this, we introduce three additional loss components as below.

1) *Ensuring Representativeness Via Similarity Alignment*: To ensure that the selected prompts $\mathcal{P}_t$ are representative of the current target-domain graph G_t , we perform clustering on node embeddings. Specifically, given the incoming graph G_t with node features $X_t \in \mathbb{R}^{N_t \times d_x}$, we first obtain backbone-induced node embeddings via the frozen backbone:

$$Z_t = f(\Theta_0)(G_t) \in \mathbb{R}^{N_t \times d_h}, \quad (4)$$

and apply K-means clustering on Z_t to partition the embeddings into M clusters:

$$\{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_M\} = \text{KMeans}(Z_t, M), \quad (5)$$

where $\mathbf{c}_i \in \mathbb{R}^{d_h}$ denotes the centroid of the i -th cluster. Each centroid $\mathbf{c}_i$ is weighted by the relative size of its cluster $\mathcal{C}_i$, and the weighted sum yields a single representative prototype embedding $\mathbf{h}$:

$$\mathbf{h} = \sum_{i=1}^M \frac{|\mathcal{C}_i|}{N_t} \mathbf{c}_i. \quad (6)$$

To align the selected prompts with this representative embedding, we compute the similarity loss $\mathcal{L}_{Ers}$ between the normalized prototype $\mathbf{h}$ and the average of the normalized prompt vectors $\{\mathbf{p}_1, \dots, \mathbf{p}_{k_p}\}$:

$$\mathcal{L}_{Ers} = \left\| \frac{\mathbf{h}}{\|\mathbf{h}\|_2} - \frac{1}{k_p} \sum_{j=1}^{k_p} \frac{\mathbf{p}_j}{\|\mathbf{p}_j\|_2} \right\|_2^2. \quad (7)$$

By minimizing $\mathcal{L}_{Ers}$, the prompt representations are explicitly aligned with the representative embedding derived from node clusters. This ensures that the top- k_p prompts effectively capture the dominant structural and semantic information of G_t , thereby facilitating reliable knowledge transfer to subsequent domains and complementing the KL divergence regularization $\mathcal{L}_{KLd}$ for forgetting mitigation.

2) *Enhancing Generalization Via Diversity Constraints*: To enhance generalization, we introduce a diversity constraint that penalizes redundancy among the selected prompts in the current training session. Given the normalized selected prompts in the embedding space $\{\tilde{\mathbf{p}}_i\}_{i=1}^{k_p} \subset \mathbb{R}^{d_h}$, we construct the Gram matrix $\mathbf{G} \in \mathbb{R}^{k_p \times k_p}$, where each element is defined as the cosine similarity between prompts:

$$G_{mn} = \tilde{\mathbf{p}}_m^\top \tilde{\mathbf{p}}_n = \cos(\tilde{\mathbf{p}}_m, \tilde{\mathbf{p}}_n), \quad m, n \in \{1, 2, \dots, k_p\}, \quad (8)$$

Then the diversity constraint is formulated as:

$$\mathcal{L}_{Div} = \frac{2}{k_p(k_p - 1)} \sum_{1 \leq m < n \leq k_p} \cos(\tilde{\mathbf{p}}_m, \tilde{\mathbf{p}}_n). \quad (9)$$

Minimizing $\mathcal{L}_{Div}$ reduces the pairwise cosine similarities among the selected prompts, effectively encouraging them to explore diverse directions in the embedding space. This promotes the model's ability to generalize across various patterns in the graph and mitigates overfitting to specific structures. It also enables the prompts to capture a broader range of information. By fostering diversity among the prompts, the model is encouraged to explore novel, rarely encountered aspects of the data, thereby enhancing its generalization capabilities. The ablation experiments prove the critical role of $\mathcal{L}_{Div}$.

3) *Mitigating Forgetting Via KL Divergence Regularization*: To mitigate catastrophic forgetting, we employ a knowledge distillation approach using the Kullback-Leibler divergence between the outputs of the current model and a saved version of the model (previous adaptation). The saved logits, which include

the prompt information, are updated using Exponential Moving Average (EMA) [39] to incorporate information from previous graphs while preserving stability.

Let $\mathbf{l}_{t-1}^{prev}, \mathbf{l}_t^{current} \in \mathbb{R}^C$ represent the average logits (including prompt information) for the previous and current steps, respectively. First, the current logits $\mathbf{l}_t^{current}$ are computed as:

$$\mathbf{l}_t^{current} = \frac{1}{N_t'} \sum_{v' \in V_t'} \ell_{v'} \in \mathbb{R}^C, \quad (10)$$

where $\ell_{v'} \in \mathbb{R}^C$ denotes the output logits for each node $v' \in V_t'$ from $f(\Theta_0)$. Since the base model is frozen during adaptation, for each class $c = 1, \dots, C$, we compute the probability distributions for the previous and current steps using the softmax function as follows: $p_{t-1,c}^{prev} = \frac{\exp(\mathbf{l}_{t-1,c}^{prev}/T)}{\sum_{c'=1}^C \exp(\mathbf{l}_{t-1,c'}^{prev}/T)}$ and $p_{t,c}^{current} = \frac{\exp(\mathbf{l}_{t,c}^{current}/T)}{\sum_{c'=1}^C \exp(\mathbf{l}_{t,c'}^{current}/T)}$. Then $\mathcal{L}_{KLd}$ is defined as:

$$\mathcal{L}_{KLd} = D_{KL}(\mathbf{p}_{t-1}^{prev} \parallel \mathbf{p}_t^{current}) = \sum_{c=1}^C p_{t-1,c}^{prev} \log \left(\frac{p_{t-1,c}^{prev}}{p_{t,c}^{current}} \right). \quad (11)$$

Minimizing $\mathcal{L}_{KLd}$ aligns the current model's predictions with those of the EMA-updated saved logits, retaining prior knowledge and mitigating forgetting by distilling information from previous graphs into the current prompts. Specifically, after training on each graph, we update the saved logits $\mathbf{l}_t^{prev}$ using EMA:

$$\mathbf{l}_t^{prev} \leftarrow \alpha \mathbf{l}_t^{current} + (1 - \alpha) \mathbf{l}_{t-1}^{prev}, \quad (12)$$

where α denotes the EMA decay rate, which we set to 0.9 following common practice for smoothing updates in continual learning. At the first adaptation step, we directly initialize $\mathbf{l}_1^{prev}$ with $\mathbf{l}_1^{current}$.

4) *Overall Objective Function*: Combining the adaptation loss with the three additional loss components, the overall loss function for the t -th adaptation step is:

$$\min_{\mathcal{P}_t} \mathcal{L}_{DPCGL}(\mathcal{G}_t', \mathcal{P}_t, G_t; \Theta_0) = \mathcal{L}_{Adp}(\mathcal{G}_t') + \lambda_1 \mathcal{L}_{Ers}(\mathcal{P}_t, G_t) + \lambda_2 \mathcal{L}_{Div}(\mathcal{P}_t) + \lambda_3 \mathcal{L}_{KLd}(\mathcal{G}_t', \mathcal{G}_{t-1}') \quad \text{s.t. } \mathcal{P}_t \subseteq \mathcal{P}, \quad (13)$$

where λ_1, λ_2 , and λ_3 are hyperparameters used to balance the relative importance of each loss component. By jointly optimizing this objective function, we adapt the prompts to the new graph G_t while ensuring that the prompts are representative of the current data, retaining prior knowledge, and promoting diversity among the prompts. This enables continual learning in an online fashion without retraining the model parameters Θ_0 . To further illustrate the efficiency of our approach, we note that in practice we typically have $k_p \ll N_t$. The prompt parameters are optimized in the backbone-induced embedding space $\mathbb{R}^{d_h}$ (with d_h matching the hidden dimension of the frozen GNN), and are projected to $\mathbb{R}^{d_x}$ via ϕ only for node-attribute injection. As a result, the number of trainable parameters is $\mathcal{O}(k_p d_h)$, making the method lightweight compared with buffered replay schemes.

E. Theoretical Analysis

The theoretical foundation of our framework is established by analyzing prompt-induced representations under a frozen encoder $f(\Theta_0)$. Let $f(\Theta_0) : \text{graph} \rightarrow \mathbb{R}^{d_h}$ denote a pre-trained backbone. We maintain the trainable prompt memory as $\mathcal{P}_t \in \mathbb{R}^{k_p \times d_x}$. To inject prompts as additional nodes, we apply a differentiable projection $\phi : \mathbb{R}^{d_h} \rightarrow \mathbb{R}^{d_x}$ and obtain $\mathcal{P}_t' = \phi(\mathcal{P}_t) \in \mathbb{R}^{k_p \times d_x}$. Accordingly, we construct the augmented graph $\mathcal{G}_t' := G_t \cup \mathcal{G}_t$, where $\mathcal{G}_t = (\mathcal{P}_t', \mathcal{E}_t)$ denotes the prompt graph at step t . The ideal latent representation is defined as the backbone output paired with an optimal prompt set $\mathcal{P}_t^*$, i.e., $C_t^* \in \mathbb{R}^{d_h}$ where $C_t^* := f(\Theta_0)(\mathcal{G}_t^*)$ and $\mathcal{G}_t^* := G_t \cup \mathcal{G}_t$ with the optimal prompt set $\mathcal{P}_t^*$ injected. Vector- and graph-level norms are denoted by $\|\cdot\|_2$ and $\|\cdot\|_F$, respectively. For dimensional consistency, we measure graph perturbations on the pair (A_t, X_t) and define $\|\mathcal{G} - \mathcal{G}'\|_F := \sqrt{\|A - A'\|_F^2 + \|X - X'\|_F^2}$, which is compatible with the Frobenius norm used in our Lipschitz definition below. Accordingly, we define the Lipschitz modulus of the frozen backbone as

$$L_f := \sup_{\mathcal{G} \neq \mathcal{G}'} \frac{\|f(\Theta_0)(\mathcal{G}) - f(\Theta_0)(\mathcal{G}')\|_2}{\|\mathcal{G} - \mathcal{G}'\|_F}.$$

Proposition 1 (Exact prompt alignment): If the last linear layer of $f(\Theta_0)$ has full row rank d_h , there exists $\mathcal{P}_t^*$ for every domain t such that

$$f(\Theta_0)(\mathcal{G}_t^*) = C_t^*,$$

as proved in [19]. This establishes the existence of an optimal prompt for perfect alignment and anchors the definition of the ideal target C_t^* used in subsequent error bounds.

Building upon Proposition 1, we provide a quantitative bound characterizing the cumulative error during continual prompt optimization.

Theorem 1 (Cumulative RMSE control under continual prompt optimization): Let P_k be the orthogonal projector onto a k -dimensional prompt subspace. For any prompt parameters $\mathcal{P}_t'$ and the corresponding augmented graph $\mathcal{G}_t'$, the step-wise alignment error satisfies

$$\begin{aligned} \epsilon_t &:= \|f(\Theta_0)(\mathcal{G}_t') - f(\Theta_0)(\mathcal{G}_t^*)\|_2 \\ &\leq \|(I - P_k)C_t^*\|_2 + L_f \|\mathcal{G}_t' - \mathcal{G}_t^*\|_F, \end{aligned} \quad (14)$$

where $C_t^* := f(\Theta_0)(\mathcal{G}_t^*)$ and L_f is the Lipschitz constant of the encoder.

Let $S = [C_1^* \dots C_m^*]$ and $\hat{S} = [f(\Theta_0)(\mathcal{G}_1') \dots f(\Theta_0)(\mathcal{G}_m')]$. If the optimizer is γ -approximate, i.e.,

$$\|\hat{S} - S_k\|_F^2 \leq \gamma \|S - S_k\|_F^2,$$

with S_k the best rank- k approximation of S , then the continual RMSE satisfies

$$\begin{aligned} \sqrt{\frac{1}{m} \sum_{t=1}^m \epsilon_t^2} &\leq \min \left\{ \sqrt{2} \left(\frac{1}{\sqrt{m}} \|(I - P_k)S\|_F + L_f \Delta_{\text{rms}} \right), \right. \\ &\quad \left. (1 + \sqrt{\gamma}) \sqrt{\lambda_{k+1}} \right\}, \end{aligned} \quad (15)$$

where

$$\Delta_{\text{rms}} := \sqrt{\frac{1}{m} \sum_{t=1}^m \|\mathcal{G}'_t - \mathcal{G}^*_t\|_F^2}, \quad \lambda_{k+1} := \frac{1}{m} \sum_{i=k+1}^r \sigma_i^2,$$

and $\{\sigma_i\}$ are the singular values of S with $r = \text{rank}(S)$. Note that $\|S - S_k\|_F^2 = \sum_{i=k+1}^r \sigma_i^2 = m \lambda_{k+1}$ by the Eckart–Young–Mirsky theorem. Here k denotes the rank of the prompt subspace (distinct from the prompt selection size k_p).

Proof: Let $S = [C_1^* \dots C_m^*] \in \mathbb{R}^{d_h \times m}$ and S_k be its best rank- k approximation. Define $\hat{S} = [f(\Theta_0)(\mathcal{G}'_1) \dots f(\Theta_0)(\mathcal{G}'_m)]$, and assume the optimizer is γ -approximate: $\|\hat{S} - S_k\|_F^2 \leq \gamma \|S - S_k\|_F^2$.

Step 1. Single-step decomposition: For any t , by the triangle inequality and the definition of L_f , we have

$$\begin{aligned} \epsilon_t &:= \|f(\Theta_0)(\mathcal{G}'_t) - C_t^*\|_2 \\ &\leq \|(I - P_k)C_t^*\|_2 + L_f \|\mathcal{G}'_t - \mathcal{G}_t^*\|_F. \end{aligned} \quad (16)$$

The first term $\|(I - P_k)C_t^*\|_2$ captures the spectral residual outside the k -dimensional prompt subspace, while the second term measures the deviation induced by the constructed augmented graph.

Step 2. Squared accumulation (dimensionally verifiable): Let $a_t := \|(I - P_k)C_t^*\|_2$ and $b_t := L_f \|\mathcal{G}'_t - \mathcal{G}_t^*\|_F$. Using $(a_t + b_t)^2 \leq 2a_t^2 + 2b_t^2$, we obtain

$$\epsilon_t^2 \leq 2\|(I - P_k)C_t^*\|_2^2 + 2L_f^2 \|\mathcal{G}'_t - \mathcal{G}_t^*\|_F^2. \quad (17)$$

Summing over $t = 1, \dots, m$ gives

$$\sum_{t=1}^m \epsilon_t^2 \leq 2 \sum_{t=1}^m \|(I - P_k)C_t^*\|_2^2 + 2L_f^2 \sum_{t=1}^m \|\mathcal{G}'_t - \mathcal{G}_t^*\|_F^2. \quad (18)$$

Step 3. RMSE upper bound (data-driven residual branch): Dividing (18) by m and taking square roots yields

$$\begin{aligned} \sqrt{\frac{1}{m} \sum_{t=1}^m \epsilon_t^2} &\leq \sqrt{\frac{2}{m} \sum_{t=1}^m \|(I - P_k)C_t^*\|_2^2} \\ &\quad + \sqrt{\frac{2L_f^2}{m} \sum_{t=1}^m \|\mathcal{G}'_t - \mathcal{G}_t^*\|_F^2}. \end{aligned} \quad (19)$$

Noting that $\sum_{t=1}^m \|(I - P_k)C_t^*\|_2^2 = \|(I - P_k)S\|_F^2$, we obtain the first branch in (15):

$$\sqrt{\frac{1}{m} \sum_{t=1}^m \epsilon_t^2} \leq \sqrt{2} \left(\frac{1}{\sqrt{m}} \|(I - P_k)S\|_F + L_f \Delta_{\text{rms}} \right), \quad (20)$$

where $\Delta_{\text{rms}} := \sqrt{\frac{1}{m} \sum_{t=1}^m \|\mathcal{G}'_t - \mathcal{G}_t^*\|_F^2}$.

Step 4. Spectral-tail branch via rank- k approximation (dimensionally verifiable): By the Eckart–Young–Mirsky theorem, $\|S - S_k\|_F^2 = \sum_{i=k+1}^r \sigma_i^2$. Under the γ -approximate optimizer condition, $\|\hat{S} - S_k\|_F \leq \sqrt{\gamma} \|S - S_k\|_F$. Then, by the triangle inequality,

$$\|\hat{S} - S\|_F \leq \|\hat{S} - S_k\|_F + \|S - S_k\|_F \leq (1 + \sqrt{\gamma}) \|S - S_k\|_F. \quad (21)$$

Dividing by $\sqrt{m}$ and using $\lambda_{k+1} = \frac{1}{m} \|S - S_k\|_F^2$ gives

$$\frac{1}{\sqrt{m}} \|\hat{S} - S\|_F \leq (1 + \sqrt{\gamma}) \sqrt{\lambda_{k+1}}, \quad (22)$$

which yields the second (spectral-tail) branch in (15).

Final unified bound: Since (20) and (22) provide two valid upper bounds on the same RMSE quantity, we take the tighter one and obtain exactly (15). In particular, if $\text{rank}(S) \leq k$ (i.e., $\lambda_{k+1} = 0$), the spectral-tail branch vanishes; if additionally $\Delta_{\text{rms}} \rightarrow 0$, the data-driven residual also vanishes, yielding near-zero RMSE without updating the frozen backbone.

Discussion and Optimization: Theorem 1 provides a dual-bound characterization of continual adaptation, consisting of a data-driven residual and a spectral-tail term. The former captures the empirical deviation from constructed prompt–graph pairs, while the latter reflects the inherent structural limit of oracle representations. Together, they reveal that, under suitable prompt ratios and subspace alignment, prompt-based test-time tuning with a frozen backbone achieves provably stable continual adaptation. As formulated in (13), the overall loss jointly regulates adaptation, alignment, and stability, and the prompt parameters are updated via stochastic gradient descent (SGD) as

$$\mathcal{P}_t \leftarrow \mathcal{P}_t - \eta \left(\frac{\partial \mathcal{L}_{\text{Adp}}}{\partial \mathcal{P}_t} + \lambda_1 \frac{\partial \mathcal{L}_{\text{Ers}}}{\partial \mathcal{P}_t} + \lambda_2 \frac{\partial \mathcal{L}_{\text{Div}}}{\partial \mathcal{P}_t} + \lambda_3 \frac{\partial \mathcal{L}_{\text{KLd}}}{\partial \mathcal{P}_t} \right),$$

where η is the learning rate and only $\mathcal{P}_t$ is optimized while the encoder $f(\Theta_0)$ is frozen. Each component corresponds to a specific part of the theoretical bound: $\mathcal{L}_{\text{Adp}}$ maintains predictive accuracy, $\mathcal{L}_{\text{Ers}}$ minimizes spectral residuals, $\mathcal{L}_{\text{KLd}}$ stabilizes inter-domain transitions, and $\mathcal{L}_{\text{Div}}$ promotes expressive prompts. Together, these synchronized objectives ground the theoretical guarantees of Theorem 1 in practical optimization.

F. Complexity Analysis

1) Time Complexity Analysis: The time complexity of DPCGL comprises prompt selection, sparse prompt-graph augmentation, and test-time adaptation/inference with a frozen backbone. We denote the dominant cost of one frozen-backbone forward–backward pass on a graph G_t as $\mathcal{C}(G_t)$ (typically linear in graph size for message-passing GNNs, e.g., $\mathcal{C}(G_t) = \mathcal{O}(L(|\mathcal{E}(G_t)| d_h + N_t d_h))$ up to implementation details). After computing the backbone-induced embeddings (accounted for by $\mathcal{C}(G_t)$), ranking prompts by cosine similarity between $\mathbf{z}_t$ and $\{\mathbf{p}_j\}_{j=1}^{|\mathcal{P}|}$ costs $\mathcal{O}(|\mathcal{P}| \cdot d_h)$. Constructing the augmented graph $\mathcal{G}'_t$ further requires computing prompt–node similarities in the embedding space, whose dominant term is $\mathcal{O}(k_p N_t d_h)$ since $k_p \ll N_t$ (other terms such as prompt–prompt similarities are lower-order). During test-time adaptation, although only the selected prompts are updated, computing their gradients still requires one backbone forward–backward pass on $\mathcal{G}'_t$, so the dominant adaptation cost is $\mathcal{C}(\mathcal{G}'_t)$. Moreover, because the augmentation is sparse and $k_p \ll N_t$, we have $\mathcal{C}(\mathcal{G}'_t) \approx \mathcal{C}(G_t)$ in practice. Overall, the per-domain overhead for selection and edge construction is $\mathcal{O}(|\mathcal{P}| d_h + k_p N_t d_h)$, plus the dominant frozen-backbone cost $\mathcal{C}(\mathcal{G}'_t)$ for test-time adaptation/inference.

TABLE II
STATISTICS OF DATASETS WITH DISTRIBUTION SHIFTS

Category	Dataset	#Nodes	#Edges	#Domains
Cross-Domain Transfers	Twitch-explicit	1,912–9,498	31,299–153,138	7
	Facebook-100	769–41,554	33,312–2,724,458	12
Temporal Shifts	Elliptic	1,089–7,880	1,168–9,164	41
	OGB-Arxiv	4,427–39,711	1,225–38,735	11

2) *Space Complexity Analysis*: The prompt pool stores $|\mathcal{P}|$ prompt embeddings, yielding space $\mathcal{O}(|\mathcal{P}|d_h)$, where d_h is the embedding dimension used for prompt optimization. At adaptation time, only k_p selected prompts are trainable, contributing $\mathcal{O}(k_p d_h)$ trainable parameters, plus sparse prompt–node edges (e.g., $\mathcal{O}(k_p k_n)$) for constructing $\mathcal{G}'_t$. For comparison, a typical GNN backbone contains $\mathcal{O}(|\Theta_0|)$ parameters (e.g., $\mathcal{O}(d_{in}d_h + (L-1)d_h^2)$ in a simplified form). The GTrans baseline introduces feature perturbation $\Delta X \in \mathbb{R}^{N \times d_x}$ and structure perturbation $\Delta A \in \{0, 1\}^{|\mathcal{E}|}$, leading to space $\mathcal{O}(Nd_x + |\mathcal{E}|)$. In contrast, our framework only stores prompt embeddings and sparse prompt connections; since $k_p \ll N_t$, the overall memory footprint remains lightweight while maintaining competitive performance.

IV. EXPERIMENTS

A. Experimental Setup

1) *Datasets*: We evaluate DPCGL on two main types of datasets: Cross-Domain Transfers and Temporal Shifts. Comprehensive dataset statistics are summarized in Table II.

The Facebook-100 [51] collection includes social networks from different U.S. universities in 2005, exhibiting notable variations in distribution and density. Twitch-Explicit [52] includes regional networks of Twitch users, where edges indicate mutual friendships among users from different regions. OGB-Arxiv [53] is composed of computer science papers from 40 subject areas on Arxiv, reflecting evolving citation patterns over time. Elliptic [54] models Bitcoin transaction networks, with nodes denoting transactions and edges describing payment links. Roughly 20% of these transactions are well labeled, supporting the identification of illicit behaviors under temporal distribution shifts.

2) *Baselines*: We compare DPCGL with a wide range of baseline methods, grouped according to their underlying domain adaptation strategies. The first group includes Test and DANN [55]. The second group refers to "Test-Time Adaptation Without Rehearsal", includes methods such as Tent [56], BN Stats Adapt [57], and EERM [58]. The third group mainly focuses on *graph-specific* test-time adaptation and parameter-efficient methods that explicitly target graph distribution shifts, including GTrans [35], GraphCTA [36], GraphATA [37], Matcha [38], TPP [49], and GraphLoRA [59]. Finally, "Continual Test-Time Adaptation", includes CoTTA [39], EATA [40], EcoTTA [41] and BECoTTA [42]. For approaches that were not initially developed for graph data, we employ GCN as the backbone architecture for fair comparison. An exception is made

for the OGB-Arxiv dataset, where GraphSAGE is adopted due to its stronger performance on large-scale citation graphs.

3) *Experimental Settings*: The GNN model was pretrained on three university graphs of the Facebook-100: Caltech36, Amherst41 and Johns Hopkins55. Our approach was then continually applied to the remaining 11 graphs. For the Twitch-explicit dataset, pretraining was done on the DE network, followed by test-time adaptation to remaining six regional networks. For the OGB-Arxiv dataset, pretraining used data before 2011, with test-time adaptation on data from after 2011. On the Elliptic dataset, we pre-trained on snapshots 7 to 9, due to the class imbalance of first six graphs, and continued training on remaining snapshots, capturing dynamic financial patterns. Our training strategy involves pretraining on selected graphs for each dataset, followed by online adaptation of prompts and inference on remaining datasets all without labels.

4) *Evaluation Metric*: We assess performance using an accuracy matrix $M^{acc} \in \mathbb{R}^{T \times T}$, where each element $M_{i,j}^{acc}$ (for $i \geq j$) indicates the accuracy on domain j after training on domain i . The Average Performance (AP) is defined as $\frac{1}{T} \sum_{i=1}^T M_{T,i}^{acc}$, which measures the model's adaptation ability, while Average Forgetting (AF) is computed as $\frac{1}{T-1} \sum_{i=1}^{T-1} (M_{T,i}^{acc} - M_{i,i}^{acc})$ to evaluate how well the model retains previous knowledge. For evaluation metrics, we adopt ROC-AUC for Twitch-Explicit, Accuracy for Facebook-100 and OGB-Arxiv, and F1 score for Elliptic, following common practice in graph distribution shift settings [58]. Under this convention, higher AF is better: $AF < 0$ indicates forgetting, while $AF > 0$ reflects positive backward transfer.

B. Experimental Results

1) *Overall Performance Comparison.(RQ1)*: We compare DPCGL with a comprehensive set of baselines categorized by different domain adaptation strategies, with results summarized in Table III. DPCGL establishes a new sota, surpassing all baselines across all datasets. Notably, certain baselines like TENT, even with adaptation strategies, exhibit poor performance due to the challenging nature of continual adaptation across multiple time steps. Among advanced baselines, CoTTA is the most comparable to DPCGL. Instead of solely relying on EMA for retaining past logits, DPCGL ensures the representativeness of previous prompts to enhance knowledge transfer. By incorporating multiple complementary loss components, our strategy leverages previous insights for improved prompt adaptation, enriching the training process while improving adaptability across changing domains. DANN outperforms other baselines by leveraging the source graph for adaptation, while other methods rely solely on target unlabeled data. However, DPCGL achieves superior results to DANN on all four datasets, particularly in the AP and AF metrics. This advantage stems from its capability to adapt prompts dynamically to new target domains while effectively retaining and transferring knowledge learned from previous ones.

2) *Comprehensive Review of Continuous Performance.(RQ2)*: To provide a comprehensive view of the model's performance during continual test-time training, we evaluated the

TABLE III

PERFORMANCE COMPARISON ACROSS FOUR DATASETS BETWEEN BASELINE METHODS AND THE PROPOSED FRAMEWORK DPCGL. "N/A" INDICATES THAT THE CORRESPONDING RESULT IS NOT AVAILABLE OR NOT APPLICABLE FOR THE GIVEN METHOD.

Methods	Twitch-explicit		Facebook-100		Elliptic		OGB-Arxiv	
	AP-AUC/% $\uparrow$	AF/% $\uparrow$	AP-ACC/% $\uparrow$	AF/% $\uparrow$	AP-F1/% $\uparrow$	AF/% $\uparrow$	AP-ACC/% $\uparrow$	AF/% $\uparrow$
Test	53.88 $\pm$ 0.00	N/A	50.55 $\pm$ 0.00	N/A	53.97 $\pm$ 0.00	N/A	42.43 $\pm$ 0.00	N/A
DANN	51.50 $\pm$ 0.39	0.19 $\pm$ 0.55	50.63 $\pm$ 0.64	0.38 $\pm$ 1.13	53.84 $\pm$ 0.56	-0.95 $\pm$ 0.54	42.62 $\pm$ 0.23	-1.28 $\pm$ 0.16
Norm	52.65 $\pm$ 0.00	-2.30 $\pm$ 0.00	47.62 $\pm$ 0.00	0.53 $\pm$ 0.00	54.67 $\pm$ 0.00	0.02 $\pm$ 0.00	40.21 $\pm$ 0.00	0.11 $\pm$ 0.00
TENT	52.84 $\pm$ 0.98	-1.83 $\pm$ 1.05	46.63 $\pm$ 0.00	0.61 $\pm$ 0.00	46.54 $\pm$ 0.00	0.55 $\pm$ 0.00	36.72 $\pm$ 0.37	-0.02 $\pm$ 0.18
EERM	52.08 $\pm$ 0.00	N/A	49.70 $\pm$ 0.03	N/A	46.51 $\pm$ 0.00	N/A	36.26 $\pm$ 0.00	N/A
CoTTA	53.94 $\pm$ 0.36	0.34 $\pm$ 0.41	50.12 $\pm$ 0.16	0.59 $\pm$ 0.12	54.08 $\pm$ 0.05	-1.92 $\pm$ 0.06	40.28 $\pm$ 0.01	-1.96 $\pm$ 0.01
EATA	53.56 $\pm$ 0.10	-0.72 $\pm$ 0.07	49.02 $\pm$ 3.16	-0.57 $\pm$ 0.03	50.48 $\pm$ 0.07	-0.76 $\pm$ 0.06	40.91 $\pm$ 0.70	-1.35 $\pm$ 0.59
EcoTTA	51.70 $\pm$ 0.59	-0.13 $\pm$ 1.10	49.34 $\pm$ 1.88	-0.68 $\pm$ 1.95	49.55 $\pm$ 0.92	0.57 $\pm$ 0.93	40.45 $\pm$ 1.21	-6.15 $\pm$ 1.18
BECoTTA	51.98 $\pm$ 0.34	-0.04 $\pm$ 1.11	49.47 $\pm$ 2.02	-0.58 $\pm$ 1.02	49.61 $\pm$ 0.90	0.77 $\pm$ 1.09	39.92 $\pm$ 1.30	-5.84 $\pm$ 1.26
GTrans	53.55 $\pm$ 0.29	N/A	OOM	N/A	54.25 $\pm$ 0.02	N/A	39.69 $\pm$ 0.04	N/A
GraphATA	50.53 $\pm$ 0.19	-0.29 $\pm$ 0.47	50.25 $\pm$ 2.97	0.38 $\pm$ 3.89	46.54 $\pm$ 0.02	-2.99 $\pm$ 0.55	36.43 $\pm$ 1.56	-1.14 $\pm$ 0.62
GraphCTA	50.43 $\pm$ 0.44	-0.70 $\pm$ 0.48	51.22 $\pm$ 2.85	-2.10 $\pm$ 2.10	46.82 $\pm$ 0.08	-2.21 $\pm$ 0.12	41.83 $\pm$ 0.02	-0.02 $\pm$ 0.04
Matcha	50.10 $\pm$ 1.40	-1.31 $\pm$ 1.82	47.98 $\pm$ 2.62	-0.81 $\pm$ 1.61	45.94 $\pm$ 5.18	0.02 $\pm$ 0.67	43.10 $\pm$ 0.66	-1.49 $\pm$ 1.17
TPP	50.14 $\pm$ 1.40	-0.63 $\pm$ 1.00	49.14 $\pm$ 2.97	-1.33 $\pm$ 1.53	46.57 $\pm$ 4.84	0.26 $\pm$ 0.87	42.75 $\pm$ 1.32	-0.35 $\pm$ 0.68
GraphLoRA	51.17 $\pm$ 0.84	-0.28 $\pm$ 1.29	51.38 $\pm$ 1.49	0.56 $\pm$ 1.42	49.06 $\pm$ 1.70	0.58 $\pm$ 0.39	43.21 $\pm$ 0.63	0.08 $\pm$ 0.01
DPCGL	54.92$\pm$0.09	0.43$\pm$0.10	53.30$\pm$0.03	3.44$\pm$0.11	55.12$\pm$0.03	0.72$\pm$0.46	43.31$\pm$0.70	0.12$\pm$0.23

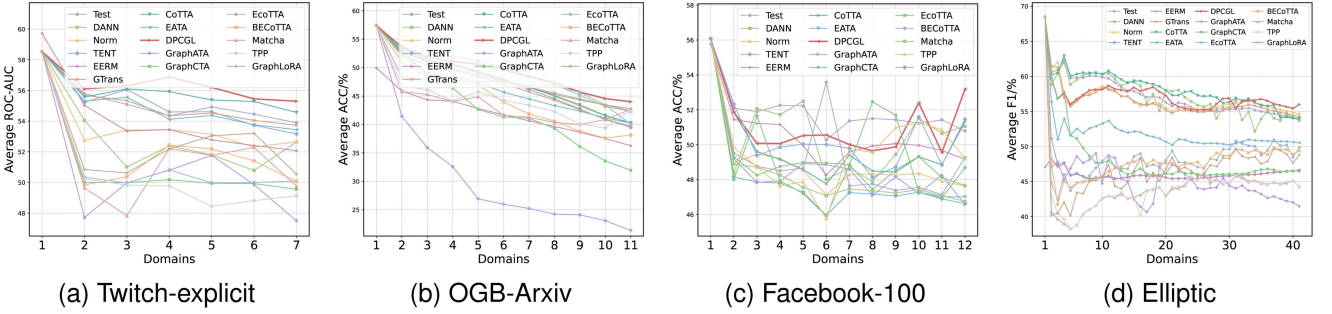


Fig. 4. Trends in average performance during continual test-time training on dynamic out-of-distribution graph domains.

average performance across all encountered domains as the prompts are updated with each new domain. Fig. 4 presents the comparative results among Test, DANN, DPCGL, and the best-performing baseline. Each curve depicts the model's performance on domain t , measured by the Average Performance (AP) over all preceding tasks. Additionally, Fig. 5 visualizes the accuracy matrices of DPCGL and CoTTA on the Twitch and OGB-Arxiv datasets for a clearer comparison of adaptation behaviors. Each row reflects the model's performance across domains as prompts are updated, while each column shows how performance evolves within a specific domain as new domains are sequentially added. Darker shades represent better performance, with lighter shades indicating worse results. As the number of domains expands, the learning objective becomes more challenging, causing a consistent decrease in performance across all compared methods. Most baselines suffer from significant performance degradation, highlighting catastrophic forgetting when previous domain data becomes inaccessible. However, DPCGL demonstrates strong resilience, consistently outperforming the comparative baseline CoTTA. In Fig. 5, DPCGL generally exhibits darker shades throughout most of the matrix compared to CoTTA, indicating better performance.

3) *Hyperparameter Experiments.(RQ3)*: With the dynamic prompt mechanism, we select k_p prompt nodes for dynamic test-time training, where k_p is much smaller than the total number of nodes in the current graph. Specifically, we set k_p as different proportions of the total number of nodes for each dataset, with slightly higher ratios used for larger graphs to ensure stable adaptation. These parameter ranges were determined based on preliminary testing to identify the most appropriate settings for each dataset. As illustrated in Fig. 6, the proposed framework consistently maintains high performance, even under a low prompt ratio setting. These observations highlight the efficiency and robustness of DPCGL, demonstrating that it can maintain stable performance even with a relatively small prompt ratio especially for Facebook-100 and Elliptic. Additionally, the experiments of loss weight sensitivity are in the Fig. 10.

4) *Ablation Studies.(RQ4)*: To this end, we designed four variant methods for DPCGL to verify the contributions of each component referred to Table IV. It can be seen that combining all the proposed loss terms with the shared prompt pool achieves the best performance, clearly indicating that each component plays a beneficial role in enhancing the effectiveness of DPCGL. When the diversity loss $\mathcal{L}_{Div}$ is removed, the model still maintains strong performance, mainly due to the complementary effects

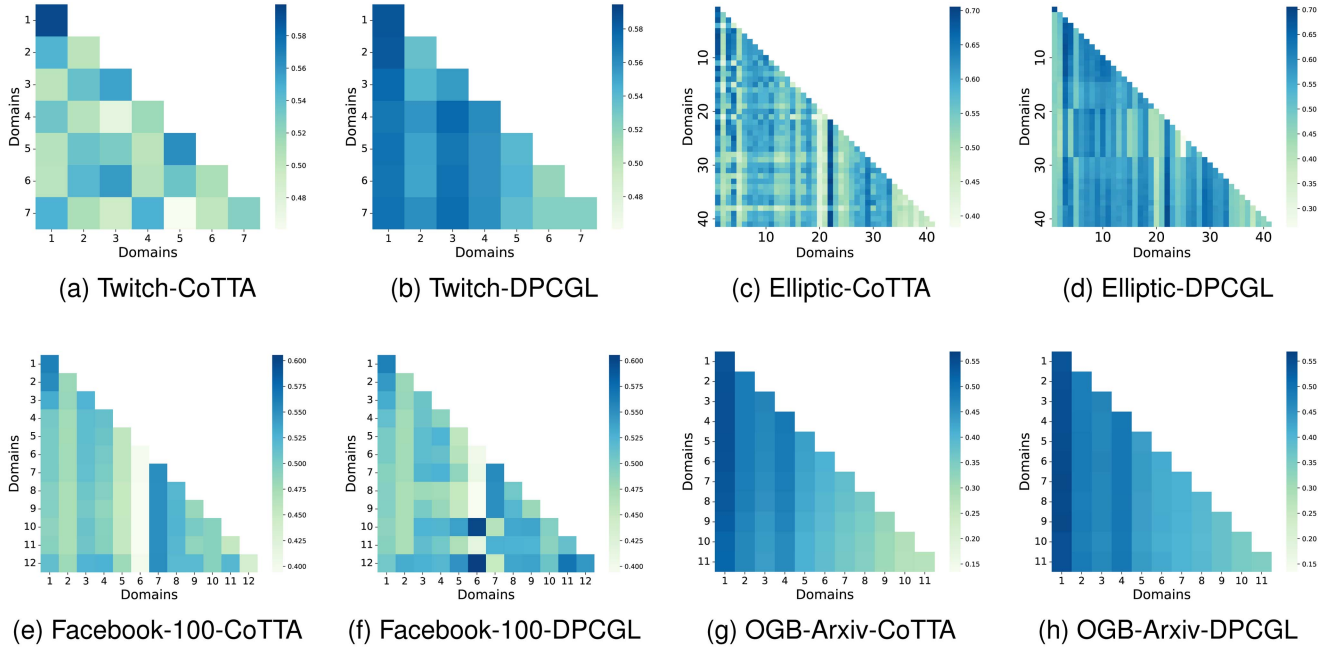


Fig. 5. Accuracy matrices of DPCGL and CoTTA across four datasets. All plots are displayed with consistent width for fair comparison.

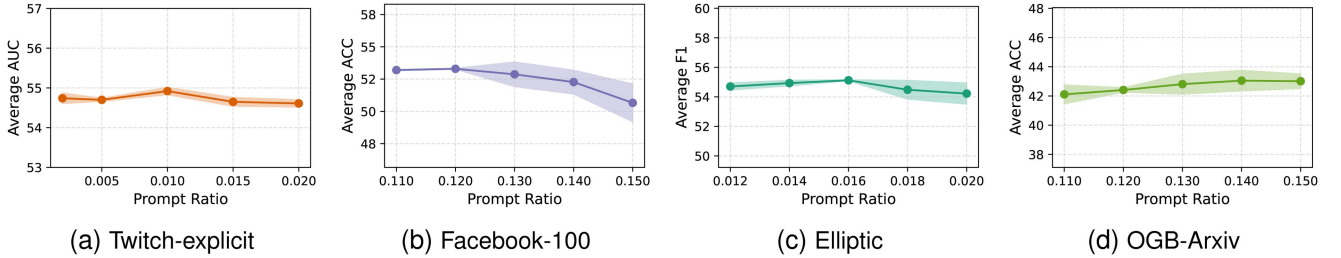


Fig. 6. Analysis of model performance across multiple datasets under different prompt ratio configurations.

TABLE IV
ABLATION STUDY OF DIFFERENT MODULES

Methods	Twitch	FB100	Elliptic	Arxiv
w/o all	53.84±0.15	52.54±0.23	54.23±0.53	32.37±0.82
w/o $\mathcal{L}_{Ers}$	54.45±0.23	53.14±0.05	54.71±0.39	41.77±0.72
w/o $\mathcal{L}_{KLd}$	54.53±0.18	52.81±0.01	54.68±0.33	35.55±3.86
w/o $\mathcal{L}_{Div}$	54.39±0.28	53.21±0.08	54.83±0.15	41.54±0.51
DPCGL	54.92±0.09	53.30±0.03	55.12±0.03	43.31±0.70

of the similarity alignment and KL divergence objectives, which is particularly noticeable on the OGB-Arxiv dataset. These two losses serve complementary purposes: $\mathcal{L}_{Ers}$ ensures that key information from the current graph is preserved, while $\mathcal{L}_{KLd}$ effectively distills and transfers this knowledge across domains. Together, they form an essential foundation for maintaining robust performance.

5) *Prompts Selection Frequency Analysis.(RQ5)*: To illustrate the frequency of prompt selection for inference after each domain of training, we visualize the selection frequency of prompt IDs across different domains in Fig. 7. The vertical axis represents the prompt IDs, while the z-axis indicates the selection frequency as a percentage. As the task(domain) ID increases, the selection of prompts becomes increasingly diverse

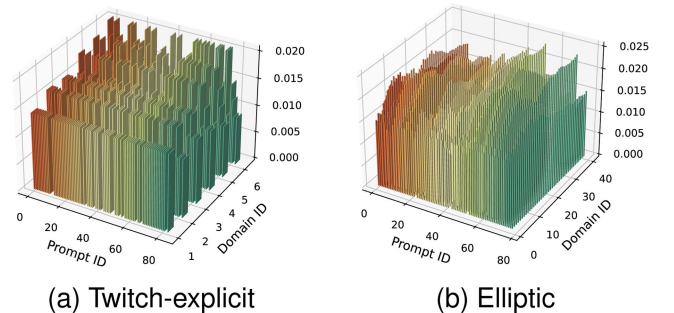


Fig. 7. Prompt selection frequency as evolving OOD domains increase.

compared to the initial stages, effectively promoting information gain. Moreover, certain prompt IDs are selected more frequently across multiple inferences, indicating a higher selection frequency. These observations collectively demonstrate the flexibility of DPCGL, as it dynamically adapts to evolving domains while retaining valuable knowledge. Furthermore, its continuous adaptive adjustments enhance information flow and promote incremental gains.

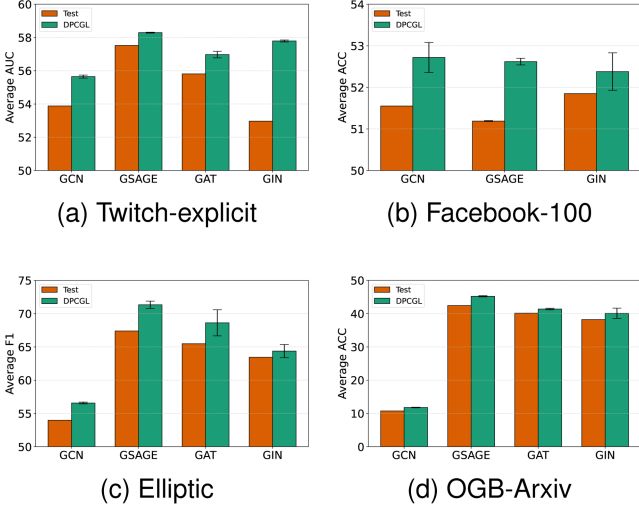


Fig. 8. Performance comparison on different GNN backbones.

C. Performance of GNN Backbones (RQ6)

We analyze the performance of different GNN architectures within our framework. Four common backbones like GCN, GraphSAGE, GAT, and GIN are selected for evaluation, as shown in Fig. 8. Each model is tested both independently and when integrated with DPCGL to assess its contribution to continual adaptation. Across all cases, DPCGL consistently improves performance over the standalone models, verifying the benefit of prompt-based adaptation. These consistent gains across architectures confirm that DPCGL is model-agnostic and generalizable to various graph backbones. Compared with the *Test* baseline that performs direct inference without adaptation, our approach achieves better stability and robustness across domains.

D. Performance of Top- k_n Parameters of Node-Prompt Edge Connection (RQ7)

As observed in the Fig. 9, the impact of increasing the top- k_n parameter varies across different datasets with fixed predefined threshold τ_s . Specifically, both the fb100 and elliptic datasets exhibit a downward trend as top- k_n increases, suggesting a negative effect as the number of edges grows. Based on experimental results, we selected a top- k_n value of 6 for the arxiv dataset and 4 for the others, demonstrating that an appropriate prompt-node connection can effectively balance performance and network complexity.

E. Loss Weights Sensitivity (RQ8)

To ensure a comprehensive evaluation, we select distinct ranges of values for each loss weight hyperparameter, considering the characteristics of different datasets. Specifically, for λ_1 , the range is set to [1, 2, 3, 4, 5] on Facebook-100 and [3, 3.5, 4, 4.5, 5] on OGB-Arxiv. For λ_2 , we adopt [1, 2, 3, 4, 5] on Facebook-100 and [0.5, 1, 2, 3, 4] on OGB-Arxiv. Finally, for λ_3 , the range is [1, 2, 3, 4, 5] on Facebook-100 and [13, 14, 15, 16, 17] on OGB-Arxiv. Note that these are the

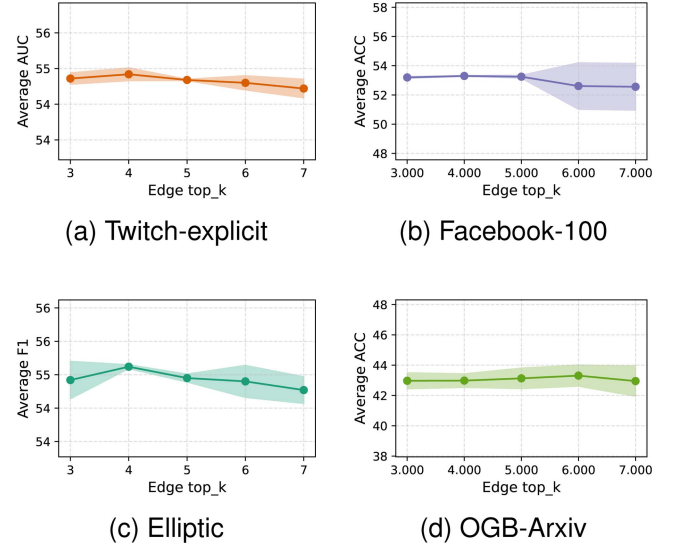
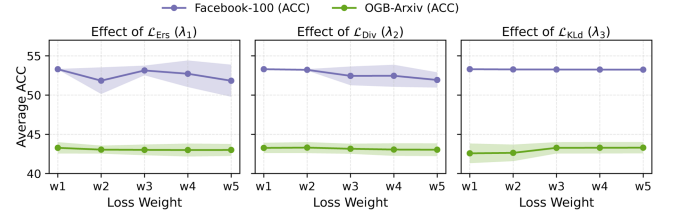
Fig. 9. Performance comparison of top- k node-prompt edge connections across datasets.

Fig. 10. Sensitivity analysis of loss weights on model performance across domains.

TABLE V
SENSITIVITY OF DPCGL TO THE PROMPT-PROMPT SIMILARITY THRESHOLD τ_s
(HIGHER IS BETTER)

Dataset	$\tau_s = 0.05$	$\tau_s = 0.10$	$\tau_s = 0.20$
Twitch-explicit	54.78 $\pm$ 0.34	54.92 $\pm$ 0.09	54.85 $\pm$ 0.16
Facebook-100	53.30 $\pm$ 0.03	52.76 $\pm$ 1.31	52.89 $\pm$ 1.07
Elliptic	55.12 $\pm$ 0.03	55.05 $\pm$ 0.16	55.04 $\pm$ 0.07
OGB-Arxiv	43.31 $\pm$ 0.70	42.91 $\pm$ 0.92	43.12 $\pm$ 0.87

raw numerical values, while the final weights are normalized as proportions for training stability. We analyze the evaluation metrics of the Facebook-100 and OGB-Arxiv datasets, which are both large-scale benchmarks, under increasing loss weights. It can be clearly observed that Facebook-100 exhibits higher sensitivity than OGB-Arxiv, particularly with respect to λ_1 and λ_2 as shown in Fig. 10. In contrast, the other two datasets display stronger robustness, which we attribute to the stability of our framework.

F. Similarity Threshold Sensitivity (RQ9)

We study the sensitivity of DPCGL to the prompt-prompt similarity threshold τ_s used for prompt graph edge construction. We evaluate three representative values $\tau_s \in \{0.05, 0.10, 0.20\}$ on all benchmarks, while keeping the prompt pool, optimization, and other hyperparameters unchanged. As shown in Table V, DPCGL achieves stable performance across different τ_s

choices, indicating that our method is not sensitive to a specific threshold and remains robust under varying prompt connectivity sparsity.

V. CONCLUSION

In conclusion, DPCGL addresses the critical challenge of continual test-time training on streaming OOD graphs. By employing a dynamic prompt-based continual test-time training framework, DPCGL effectively enables label-free adaptation, mitigates catastrophic forgetting, and offers an adaptable, efficient solution for scalable continual graph adaptation. Comprehensive theoretical analysis together with extensive empirical validation demonstrates that the proposed flexible framework consistently outperforms existing approaches, particularly in terms of adaptability and knowledge preservation.

REFERENCES

- [1] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. Int. Conf. Learn. Representations*, 2017.
- [2] W. Hamilton, Z. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017.
- [3] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Lio, and Y. Bengio, "Graph attention networks," 2017, *arXiv:1710.10903*.
- [4] L. Tang and H. Liu, "Graph mining applications to social network analysis," in *Managing Mining Graph Data*. Berlin, Germany: Springer, 2010, pp. 487–513.
- [5] X. Sun et al., "Multi-level hyperedge distillation for social linking prediction on sparsely observed networks," in *Proc. Web Conf.*, 2021, pp. 2934–2945.
- [6] Z. Gao et al., "Hierarchical graph learning for protein–protein interaction," *Nature Commun.*, vol. 14, no. 1, 2023, Art. no. 1093.
- [7] X. Zhang, C. Xu, X. Tian, and D. Tao, "Graph edge convolutional neural networks for skeleton-based action recognition," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 8, pp. 3047–3060, Aug. 2020.
- [8] X. He, K. Deng, X. Wang, Y. Li, Y. Zhang, and M. Wang, "LightGCN: Simplifying and powering graph convolution network for recommendation," in *Proc. 43rd Int. ACM SIGIR Conf. Res. Develop. Inf. Retrieval*, 2020, pp. 639–648.
- [9] S. Gui, M. Liu, X. Li, Y. Luo, and S. Ji, "Joint learning of label and environment causal independence for graph out-of-distribution generalization," in *Proc. Adv. Neural Inf. Process. Syst.*, 2023, pp. 3945–3978.
- [10] X. Li, S. Gui, Y. Luo, and S. Ji, "Graph structure extrapolation for out-of-distribution generalization," in *Proc. 41st Int. Conf. Mach. Learn.*, 2024, pp. 27846–27874.
- [11] Z. Qiao et al., "GCAL: Adapting graph models to evolving domain shifts," in *Proc. 42nd Int. Conf. Mach. Learn.*, 2025, pp. 50113–50129.
- [12] X. Zhang, D. Song, and D. Tao, "Sparsified subgraph memory for continual graph representation learning," in *Proc. IEEE Int. Conf. Data Mining*, 2022, pp. 1335–1340.
- [13] R. Aljundi, K. Kelchtermans, and T. Tuytelaars, "Task-free continual learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 11254–11263.
- [14] Z. Wang et al., "Learning to prompt for continual learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 139–149.
- [15] Z. Qiao, J. Xiao, Q. Sun, M. Xiao, X. Luo, and H. Xiong, "Towards continuous reuse of graph models via holistic memory diversification," in *Proc. 13th Int. Conf. Learn. Representations*, 2025.
- [16] J. Li et al., "Can large language models help experimental design for causal discovery?," in *Proc. ICML 2025 Workshop Scaling Intervention Models*, 2025.
- [17] R. Cai, C. Wang, Q. Cai, D. Shen, and H. Xiong, "Boosting knowledge graph-based recommendations through confidence-aware augmentation with large language models," 2025, *arXiv:2502.03715*.
- [18] L. Xu, H. Xie, S. J. Qin, X. Tao, and F. L. Wang, "Parameter-efficient fine-tuning methods for pretrained language models: A critical review and assessment," *IEEE Trans. Pattern Anal. Mach. Intell.*, pp. 1–20, 2026.
- [19] Q. Wang, X. Sun, and H. Cheng, "Does graph prompt work? A data operation perspective with theoretical analysis," in *Proc. 42nd Int. Conf. Mach. Learn.*, 2025, pp. 64377–64402.
- [20] F. Zhou and C. Cao, "Overcoming catastrophic forgetting in graph neural networks with experience replay," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 4714–4722.
- [21] Y. Liu, R. Qiu, and Z. Huang, "CAT: Balanced continual graph learning with graph condensation," in *Proc. IEEE Int. Conf. Data Mining*, 2023, pp. 1157–1162.
- [22] D. Li, A. Zhang, J. Gao, and B. Qi, "An efficient memory module for graph few-shot class-incremental learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 130084–130108.
- [23] H. Liu, Y. Yang, and X. Wang, "Overcoming catastrophic forgetting in graph neural networks," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 8653–8661.
- [24] L. Sun, J. Ye, H. Peng, F. Wang, and S. Y. Philip, "Self-supervised continual graph learning in adaptive riemannian spaces," in *Proc. AAAI Conf. Artif. Intell.*, 2023, pp. 4633–4642.
- [25] Y. Xu, Y. Zhang, W. Guo, H. Guo, R. Tang, and M. Coates, "GraphSAIL: Graph structure aware incremental learning for recommender systems," in *Proc. 29th ACM Int. Conf. Inf. Knowl. Manage.*, 2020, pp. 2861–2868.
- [26] J. Cai et al., "Multimodal continual graph learning with neural architecture search," in *Proc. ACM Web Conf.*, 2022, pp. 1292–1300.
- [27] X. Zhang, D. Song, and D. Tao, "Ricci curvature-based graph sparsification for continual graph representation learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 12, pp. 17398–17410, Dec. 2024.
- [28] X. Zhang, D. Song, and D. Tao, "Hierarchical prototype networks for continual graph representation learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 4622–4636, Apr. 2023.
- [29] X. Shen, Q. Dai, S. Mao, F.-L. Chung, and K.-S. Choi, "Network together: Node classification via cross-network deep network embedding," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 5, pp. 1935–1948, May 2021.
- [30] J. Wu, J. He, and E. Ainsworth, "Non-IID transfer learning on graphs," in *Proc. AAAI Conf. Artif. Intell.*, 2023, pp. 10342–10350.
- [31] M. Wu, S. Pan, C. Zhou, X. Chang, and X. Zhu, "Unsupervised domain adaptive graph convolutional networks," in *Proc. Web Conf.*, 2020, pp. 1457–1467.
- [32] Q. Dai, X.-M. Wu, J. Xiao, X. Shen, and D. Wang, "Graph transfer learning via adversarial domain adaptation with graph convolution," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 5, pp. 4908–4922, May 2023.
- [33] Z. Qiao, X. Luo, M. Xiao, H. Dong, Y. Zhou, and H. Xiong, "Semi-supervised domain adaptation in graph transfer learning," in *Proc. 32nd Int. Joint Conf. Artif. Intell.*, 2023, pp. 2279–2287.
- [34] R. Huang, J. Xu, X. Jiang, R. An, and Y. Yang, "Can modifying data address graph domain adaptation?," in *Proc. 30th ACM SIGKDD Conf. Knowl. Discov. Data Mining*, 2024, pp. 1131–1142.
- [35] W. Jin, T. Zhao, J. Ding, Y. Liu, J. Tang, and N. Shah, "Empowering graph representation learning with test-time graph transformation," in *Proc. 11th Int. Conf. Learn. Representations*, 2022.
- [36] Z. Zhang et al., "Collaborate to adapt: Source-free graph domain adaptation via bi-directional adaptation," in *Proc. ACM Web Conf.*, 2024, pp. 664–675.
- [37] Z. Zhang and B. He, "Aggregate to adapt: Node-centric aggregation for multi-source-free graph domain adaptation," in *Proc. ACM Web Conf.*, 2025, pp. 4420–4431.
- [38] W. Bao, Z. Zeng, Z. Liu, H. Tong, and J. He, "Matcha: Mitigating graph structure shifts with test-time adaptation," in *Proc. 13th Int. Conf. Learn. Representations*, 2024.
- [39] Q. Wang, O. Fink, L. Van Gool, and D. Dai, "Continual test-time domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 7201–7211.
- [40] S. Niu et al., "Efficient test-time model adaptation without forgetting," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 16888–16905.
- [41] J. Song, J. Lee, I. S. Kweon, and S. Choi, "EcoTTA: Memory-efficient continual test-time adaptation via self-distilled regularization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 11920–11929.
- [42] D. Lee, J. Yoon, and S. J. Hwang, "BECOTTA: Input-dependent online blending of experts for continual test-time adaptation," in *Proc. 41st Int. Conf. Mach. Learn.*, 2024, pp. 27072–27093.
- [43] Z.-Y. Zhang, Z. Xie, H. Yao, and M. Sugiyama, "Test-time adaptation in non-stationary environments via adaptive representation alignment," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 94607–94632.
- [44] M. Sun, K. Zhou, X. He, Y. Wang, and X. Wang, "GPPT: Graph pre-training and prompt tuning to generalize graph neural networks," in *Proc. 28th ACM SIGKDD Conf. Knowl. Discov. Data Mining*, 2022, pp. 1717–1727.
- [45] Z. Liu, X. Yu, Y. Fang, and X. Zhang, "GraphPrompt: Unifying pre-training and downstream tasks for graph neural networks," in *Proc. ACM Web Conf.*, 2023, pp. 417–428.
- [46] X. Sun, H. Cheng, J. Li, B. Liu, and J. Guan, "All in one: Multi-task prompting for graph neural networks," in *Proc. 29th ACM SIGKDD Conf. Knowl. Discov. Data Mining*, 2023, pp. 2120–2131.

- [47] T. Fang, Y. Zhang, Y. Yang, C. Wang, and L. Chen, "Universal prompt tuning for graph neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 52464–52489.
- [48] Q. Huang et al., "PRODIGY: Enabling in-context learning over graphs," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, vol. 36, pp. 16302–16317.
- [49] C. Niu, G. Pang, L. Chen, and B. Liu, "Replay-and-forget-free graph class-incremental learning: A task profiling and prompting approach," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 87978–88002.
- [50] J. Liang, D. Hu, and J. Feng, "Do we really need to access the source data? Source hypothesis transfer for unsupervised domain adaptation," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 6028–6039.
- [51] A. L. Traud, P. J. Mucha, and M. A. Porter, "Social structure of facebook networks," *Physica A: Statist. Mechan. its Appl.*, vol. 391, no. 16, pp. 4165–4180, 2012.
- [52] B. Rozemberczki, C. Allen, and R. Sarkar, "Multi-scale attributed node embedding," *J. Complex Netw.*, vol. 9, no. 2, 2021, Art. no. cnab014.
- [53] W. Hu et al., "Open graph benchmark: Datasets for machine learning on graphs," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 22118–22133.
- [54] A. Pareja et al., "EvolveGCN: Evolving graph convolutional networks for dynamic graphs," in *Proc. AAAI Conf. Artif. Intell.*, 2020, pp. 5363–5370.
- [55] Y. Ganin et al., "Domain-adversarial training of neural networks," *J. Mach. Learn. Res.*, vol. 17, no. 59, pp. 1–35, 2016.
- [56] D. Wang, E. Shelhamer, S. Liu, B. Olshausen, and T. Darrell, "Tent: Fully test-time adaptation by entropy minimization," in *Proc. Int. Conf. Learn. Representations*, 2020.
- [57] Y. Li, N. Wang, J. Shi, J. Liu, and X. Hou, "Revisiting batch normalization for practical domain adaptation," 2016, *arXiv:1603.04779*.
- [58] Q. Wu, H. Zhang, J. Yan, and D. Wipf, "Handling distribution shifts on graphs: An invariance perspective," in *Proc. Int. Conf. Learn. Representations*, 2022.
- [59] Z.-R. Yang, J. Han, C.-D. Wang, and H. Liu, "GraphLoRA: Structure-aware contrastive low-rank adaptation for cross-graph transfer learning," in *Proc. 31st ACM SIGKDD Conf. Knowl. Discov. Data Mining*, 2025, pp. 1785–1796.



Qianyi Cai received the MPhil degree in artificial intelligence from the Hong Kong University of Science and Technology Guangzhou, in 2025. He is currently working toward the PhD degree with the AI Thrust, Information Hub, The Hong Kong University of Science and Technology. His current research interests include graph machine learning, continual learning and large language models. He has published papers in ICML, NeurIPS, WWW and IJCV.



Ziyue Qiao (Member, IEEE) received the BS degree from Wuhan University, China, in 2017, and the PhD degree from the University of Chinese Academy of Sciences, in 2022. He is currently an assistant professor with the School of Computing and Information Technology, Great Bay University. Previously, he was a postdoc with the Hong Kong University of Science and Technology (Guangzhou) from 2022 to 2024. His research interests include data mining, graph learning, and AI for science, with a focus on graph representation/transfer learning and academic data mining.



Rui Cai received the BS degree in computer science from Xi'an Jiaotong University, China, and is currently working toward the PhD degree in computer science with the University of California, Davis. His research interests include multimodal large language models and their applications to downstream tasks such as recommendation systems and graph learning.



Huijie Liu received the B.S. and Ph.D. degrees from the University of Science and Technology of China (USTC), China, in 2019 and 2025, respectively. She is currently an assistant professor with the College of Computer Science and Software Engineering at Shenzhen University. Her current research interests include network embedding, multimodal learning, and their applications in knowledge discovery and data mining.



Junyi Li received the BEng degree in statistics from the School of Mathematics, Sun Yat-sen University, Guangzhou, China, in 2023, and the MPhil degree in artificial intelligence from the Hong Kong University of Science and Technology Guangzhou, in 2025. He is currently working toward the PhD degree with the Hong Kong University of Science and Technology. His research interests include trustworthy AI and causality.



Xiao Luo received the PhD degree from Peking University (PKU). He is currently an assistant professor with the Department of Statistics, University of Wisconsin–Madison. Previously, He was a post-doctoral researcher with the Department of Computer Science, UCLA. He is a machine learning researcher with a focus on interdisciplinary applications (AI4Science). Recently, He has been focused on LLM Post-training, LLM (Agent) Applications, and Machine Learning for Graphs and Dynamical Systems.



Hui Xiong (Fellow, IEEE) received the PhD degree in computer science from the University of Minnesota, USA. He is a chair professor, associate vice president (Knowledge Transfer) with the Hong Kong University of Science and Technology Guangzhou. His research interests include span artificial intelligence, data mining, and mobile computing. He has served on numerous organization and program committees for conferences, including as Program Co-Chair for the Industrial and Government Track for the 18th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), Program Co-Chair for the IEEE 2013 International Conference on Data Mining (ICDM), General Co-Chair for the 2015 IEEE International Conference on Data Mining (ICDM), and Program Co-Chair of the Research Track for the 2018 ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. He received several awards, such as the 2021 AAAI Best Paper Award and the 2011 IEEE ICDM Best Research Paper award. For his significant contributions to data mining and mobile computing, he was elected as a fellow of both the AAAS in 2020, later a fellow of the AAAI in 2025, and a fellow of the ACM in 2026.